10. Playfair Cipher — Encrypt the Message

CODE:

```
matrix = [
    list("MFHIK"),
    list("UNOPQ"),
    list("ZVWXY"),
    list("ELARG"),
    list("DSTBC")
]

def prepare(text):
    text = "".join(ch for ch in text.upper() if ch.isalpha())
    text = text.replace("J", "I")
    pairs = []
    i = 0

    while i < len(text):
        a = text[i]
        if i + 1 == len(text):
            b = "X"
            i += 1
        elif text[i] == text[i+1]:
            b = "X"
            i += 1
        else:
            b = text[i+1]
            i += 2
        pairs.append((a, b))

    return pairs


def encrypt(pairs):
    pos = {matrix[r][c]: (r, c)
           for r in range(5) for c in range(5)}

    result = ""

    for a, b in pairs:
        r1, c1 = pos[a]
        r2, c2 = pos[b]

        if r1 == r2:
            result += matrix[r1][(c1+1) % 5]
            result += matrix[r2][(c2+1) % 5]
        elif c1 == c2:
            result += matrix[(r1+1) % 5][c1]
            result += matrix[(r2+1) % 5][c2]
        else:
            result += matrix[r1][c2]
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            result += matrix[r2][c1]

    return result


message = "Must see you over Cadogan West. Coming at once."
pairs = prepare(message)

print("Prepared pairs:", pairs)
print("Ciphertext:", encrypt(pairs))
```

OUTPUT:

